

Convolutions Practice

1. What values are produced by the convolution below?

Image

0	1	1	0
1	0	0	1
1	1	1	1
1	0	0	1

Convolution matrix

0	0	0
0.25	0.5	0.25
0	0	0

Result

Can you suggest a name for this filter?

2. What values are produced by the convolution below?

Image

1	0.25	0	0
0.25	1	0.25	0
0	0.25	1	0.25
0	0	0.25	1

Convolution matrix

0	-1	0
-1	5	-1
0	-1	0

Result

Can you suggest a name for this filter?

3. Can you suggest a convolution matrix for the following filters?

Darken an image

Detect horizontal lines
